

Unit 22: Newton's Method — Full Solutions

Warm-up

1. $f(x) = x^2 - 2$, $f'(x) = 2x$, $x_0 = 1$.

$f(1) = 1 - 2 = -1$. $f'(1) = 2$.

$$x_1 = 1 - \frac{-1}{2} = 1 + 0.5 = 1.5$$

$x_1 = 1.5$.

2. $f(x) = x^2 - 5$, $f'(x) = 2x$, $x_0 = 2$.

$f(2) = 4 - 5 = -1$. $f'(2) = 4$.

$$x_1 = 2 - \frac{-1}{4} = 2 + 0.25 = 2.25$$

$x_1 = 2.25$.

3. $f(x) = x^3 - 2$, $f'(x) = 3x^2$, $x_0 = 1$.

$f(1) = 1 - 2 = -1$. $f'(1) = 3$.

$$x_1 = 1 - \frac{-1}{3} = 1 + \frac{1}{3} = \frac{4}{3} \approx 1.3333$$

$x_1 = \frac{4}{3} \approx 1.3333$.

4. $f(x) = x^2 - 10$, $f'(x) = 2x$, $x_0 = 3$.

$f(3) = 9 - 10 = -1$. $f'(3) = 6$.

$$x_1 = 3 - \frac{-1}{6} = 3 + \frac{1}{6} = \frac{19}{6} \approx 3.1667$$

$x_1 = \frac{19}{6} \approx 3.1667$.

5. $f(x) = x^3 - 8$, $f'(x) = 3x^2$, $x_0 = 3$.

$f(3) = 27 - 8 = 19$. $f'(3) = 27$.

$$x_1 = 3 - \frac{19}{27} = \frac{81 - 19}{27} = \frac{62}{27} \approx 2.2963$$

$x_1 = \frac{62}{27} \approx 2.2963$.

6. $f(x) = x^2 - 7$, $f'(x) = 2x$, $x_0 = 3$.

$f(3) = 9 - 7 = 2$. $f'(3) = 6$.

$$x_1 = 3 - \frac{2}{6} = 3 - \frac{1}{3} = \frac{8}{3} \approx 2.6667$$

$$x_1 = \frac{8}{3} \approx 2.6667.$$

Standard

$$7. f(x) = x^2 - 3, f'(x) = 2x, x_0 = 2.$$

$$f(2) = 4 - 3 = 1. f'(2) = 4.$$

$$x_1 = 2 - \frac{1}{4} = 1.75$$

$$f(1.75) = 3.0625 - 3 = 0.0625. f'(1.75) = 3.5.$$

$$x_2 = 1.75 - \frac{0.0625}{3.5} = 1.75 - 0.017857 \approx 1.732143$$

$$x_1 = 1.75; x_2 \approx 1.732143 \text{ (the true value of } \sqrt{3} \approx 1.732051 \text{ — very close already).}$$

$$8. f(x) = x^3 - 5, f'(x) = 3x^2, x_0 = 2.$$

$$f(2) = 8 - 5 = 3. f'(2) = 12.$$

$$x_1 = 2 - \frac{3}{12} = 2 - 0.25 = 1.75$$

$$f(1.75) = 1.75^3 - 5 = 5.359375 - 5 = 0.359375. f'(1.75) = 3(1.75)^2 = 3(3.0625) = 9.1875.$$

$$x_2 = 1.75 - \frac{0.359375}{9.1875} \approx 1.75 - 0.039122 \approx 1.710878$$

$$x_1 = 1.75; x_2 \approx 1.710878 \text{ (true value } \sqrt[3]{5} \approx 1.709976).$$

$$9. f(x) = x^2 + x - 6, f'(x) = 2x + 1, x_0 = 1.$$

$$f(1) = 1 + 1 - 6 = -4. f'(1) = 3.$$

$$x_1 = 1 - \frac{-4}{3} = 1 + \frac{4}{3} = \frac{7}{3} \approx 2.3333$$

$$f\left(\frac{7}{3}\right) = \frac{49}{9} + \frac{7}{3} - 6 = \frac{49}{9} + \frac{21}{9} - \frac{54}{9} = \frac{16}{9}. f'\left(\frac{7}{3}\right) = 2\left(\frac{7}{3}\right) + 1 = \frac{14}{3} + 1 = \frac{17}{3}.$$

$$x_2 = \frac{7}{3} - \frac{16/9}{17/3} = \frac{7}{3} - \frac{16}{9} \cdot \frac{3}{17} = \frac{7}{3} - \frac{48}{153} = \frac{7}{3} - \frac{16}{51}$$

$$\text{Convert } \frac{7}{3} \text{ to fifty-firsts: } \frac{7}{3} = \frac{119}{51}.$$

$$x_2 = \frac{119}{51} - \frac{16}{51} = \frac{103}{51} \approx 2.0196$$

$x_1 = \frac{7}{3} \approx 2.3333$; $x_2 \approx 2.0196$ (converging nicely toward the true root $x = 2$).

10. $f(x) = \cos x - x$, $f'(x) = -\sin x - 1$, $x_0 = 1$.

$f(1) = \cos(1) - 1 \approx 0.5403 - 1 = -0.4597$. $f'(1) = -\sin(1) - 1 \approx -0.8415 - 1 = -1.8415$.

$$x_1 = 1 - \frac{-0.4597}{-1.8415} = 1 - 0.2497 \approx 0.7503$$

$x_1 \approx 0.7503$ (the true fixed point is ≈ 0.739085 — already quite close after one step).

11. $f(x) = x^3 - x - 1$, $f'(x) = 3x^2 - 1$, $x_0 = 1$.

$f(1) = 1 - 1 - 1 = -1$. $f'(1) = 3 - 1 = 2$.

$$x_1 = 1 - \frac{-1}{2} = 1.5$$

$f(1.5) = 3.375 - 1.5 - 1 = 0.875$. $f'(1.5) = 3(2.25) - 1 = 5.75$.

$$x_2 = 1.5 - \frac{0.875}{5.75} \approx 1.5 - 0.152174 \approx 1.347826$$

$x_1 = 1.5$; $x_2 \approx 1.347826$ (true root ≈ 1.324718 , still converging).

12. $f(x) = x^2 - 6$, $f'(x) = 2x$, $x_0 = 2$.

$f(2) = 4 - 6 = -2$. $f'(2) = 4$.

$$x_1 = 2 - \frac{-2}{4} = 2 + 0.5 = 2.5$$

$f(2.5) = 6.25 - 6 = 0.25$. $f'(2.5) = 5$.

$$x_2 = 2.5 - \frac{0.25}{5} = 2.5 - 0.05 = 2.45$$

$x_1 = 2.5$; $x_2 = 2.45$ (true value $\sqrt{6} \approx 2.449490$ — extremely close already).

Challenge

13. $f(x) = x^4 - 2$, $f'(x) = 4x^3$, $x_0 = 1$.

$f(1) = 1 - 2 = -1$. $f'(1) = 4$.

$$x_1 = 1 - \frac{-1}{4} = 1 + 0.25 = 1.25$$

$x_1 = 1.25$.

14. $f(x) = x^4 - 2$, $f'(x) = 4x^3$, $x_0 = 1$. (Same setup as Problem 13.)

$x_1 = 1.25$ (from Problem 13).

$f(1.25) = (1.25)^4 - 2$. Compute step by step: $1.25^2 = 1.5625$; $1.25^4 = (1.5625)^2 = 2.44140625$. So $f(1.25) = 2.44140625 - 2 = 0.44140625$.

$f'(1.25) = 4(1.25)^3 = 4(1.953125) = 7.8125$.

$$x_2 = 1.25 - \frac{0.44140625}{7.8125} \approx 1.25 - 0.0565 \approx 1.1935$$

$x_1 = 1.25$; $x_2 \approx 1.1935$ (the true value of $\sqrt[4]{2} \approx 1.189207$ — converging fast).

15. $f(x) = x^5 - 3$, $f'(x) = 5x^4$, $x_0 = 1$.

$f(1) = 1 - 3 = -2$. $f'(1) = 5$.

$$x_1 = 1 - \frac{-2}{5} = 1 + 0.4 = 1.4$$

Compute $f(1.4)$: $1.4^2 = 1.96$, $1.4^3 = 2.744$, $1.4^4 = 3.8416$, $1.4^5 = 5.37824$. So $f(1.4) = 5.37824 - 3 = 2.37824$.

$f'(1.4) = 5(3.8416) = 19.208$.

$$x_2 = 1.4 - \frac{2.37824}{19.208} \approx 1.4 - 0.123807 \approx 1.276193$$

$x_1 = 1.4$; $x_2 \approx 1.276193$ (the true value of $\sqrt[5]{3} \approx 1.245731$, still converging).

16. What goes wrong if $f'(x_0) = 0$?

Looking at the formula:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

If $f'(x_0) = 0$, this requires dividing by 0, which is undefined — the formula breaks down completely and cannot produce a next guess at all.

Geometrically: a derivative of 0 means the tangent line at x_0 is perfectly horizontal. A horizontal line either never crosses the x-axis (if it's not already sitting on it) or lies exactly along the x-axis

(an extremely rare coincidence) — either way, there's no well-defined single crossing point for the method to use as its next guess. In practice, if this happens, you simply need to choose a different starting guess where the derivative isn't zero.

17. $f(x) = \sin x$, $f'(x) = \cos x$, $x_0 = 3$.

$f(3) = \sin(3) \approx 0.14112$. $f'(3) = \cos(3) \approx -0.98999$.

$$x_1 = 3 - \frac{0.14112}{-0.98999} \approx 3 + 0.14255 \approx 3.14255$$

Now for x_2 : $x_1 \approx 3.14255$ is already very close to $\pi \approx 3.14159265$.

$f(3.14255) = \sin(3.14255)$. Since $3.14255 - \pi \approx 0.00096$, and $\sin(\pi + \varepsilon) \approx -\varepsilon$ for small ε :
 $f(3.14255) \approx -0.00096$.

$f'(3.14255) = \cos(3.14255) \approx \cos(\pi + 0.00096) \approx -1$ (very close to -1).

$$x_2 = 3.14255 - \frac{-0.00096}{-1} \approx 3.14255 - 0.00096 \approx 3.14159$$

$x_1 \approx 3.14255$; $x_2 \approx 3.14159$ — remarkably close to the true value of $\pi \approx 3.14159265$ after just two iterations, showing how quickly Newton's Method can converge.

Applied

18. Approximate $\sqrt{7}$: $f(x) = x^2 - 7$, $f'(x) = 2x$, $x_0 = 2.5$.

$f(2.5) = 6.25 - 7 = -0.75$. $f'(2.5) = 5$.

$$x_1 = 2.5 - \frac{-0.75}{5} = 2.5 + 0.15 = 2.65$$

$x_1 = 2.65$ (true value $\sqrt{7} \approx 2.645751$ — very close).

19. $f(x) = x^2 - 50x + 200$, $f'(x) = 2x - 50$, $x_0 = 5$.

$f(5) = 25 - 250 + 200 = -25$. $f'(5) = 10 - 50 = -40$.

$$x_1 = 5 - \frac{-25}{-40} = 5 - 0.625 = 4.375$$

Answer: $x_1 = 4.375$ units — this approximates the lower break-even production level. (Using the quadratic formula for comparison, the exact lower root is $\frac{50 - \sqrt{1700}}{2} \approx 4.3846$ — Newton's Method already lands remarkably close after a single step.)

20. $f(x) = x^3 - 2x - 5$, $f'(x) = 3x^2 - 2$, $x_0 = 2$.

$f(2) = 8 - 4 - 5 = -1$. $f'(2) = 12 - 2 = 10$.

$$x_1 = 2 - \frac{-1}{10} = 2 + 0.1 = 2.1$$

$$f(2.1) = 9.261 - 4.2 - 5 = 0.061. \quad f'(2.1) = 3(4.41) - 2 = 13.23 - 2 = 11.23.$$

$$x_2 = 2.1 - \frac{0.061}{11.23} \approx 2.1 - 0.005432 \approx 2.094568$$

$x_1 = 2.1$; $x_2 \approx 2.094568$ (the true root is ≈ 2.0945515 — this is actually the exact historical example Isaac Newton himself first used to demonstrate the method!).

21. Approximate $\sqrt[3]{10}$: $f(x) = x^3 - 10$, $f'(x) = 3x^2$, $x_0 = 2$.

$$f(2) = 8 - 10 = -2. \quad f'(2) = 12.$$

$$x_1 = 2 - \frac{-2}{12} = 2 + \frac{1}{6} = \frac{13}{6} \approx 2.1667$$

$$x_1 = \frac{13}{6} \approx 2.1667 \text{ (true value } \sqrt[3]{10} \approx 2.15443 \text{ — already close after one step).}$$